IMPERIAL COLLEGE LONDON

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EE9-FPN1-02

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART IV: MEng and ACGI

HVDC TECHNOLOGY AND CONTROL

Monday, 4 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : B. Chaudhuri, A. Junyent-Ferre
Second Marker(s) : A. Junyent-Ferre, B. Chaudhuri

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Answer ALL Questions

Question 1 and 2 carries 25 marks each, Question 3 carries 30 marks

1. a) Use of HVDC is preferred for subsea cables connecting offshore wind farms which are located far away (> 100 km) from the onshore grid.

i) Explain why this is so.

[3]

ii) Give an example where an exception to the above was made and state why it was possible in this case.

[2]

b) The following statements relate to a point-to-point LCC HVDC link. Justify or rectify the statements with proper reasoning:

i) The current order at the rectifier should be maintained at the rated value irrespective of the AC voltage.

[3]

ii) The value of the short circuit ratio is smaller than that of the effective short circuit ratio under the rated condition.

[3]

iii) Design of the DC-side smoothing reactor involves a trade-off between multiple factors.

[3]

iv) In order to take one converter out of service for maintenance, it is enough to interrupt the firing pulses to the thyristors of that converter.

[3]

c) Considering the typical values of firing angle (α), extinction advance angle (γ) and overlap angle (μ), show that the reactive power consumed at each end of a point-to-point LCC HVDC link is within 50%-70% of the active power transmitted through the link.

[3]

d) A 2.85 GW LCC HVDC link interconnects two 275 kV AC systems. The short circuit impedances of the AC systems at the point of connection with the HVDC link are 13.5 Ω and 14.0 Ω. The minimum stipulated effective short circuit ratio (ESCR) at each end is 3.5.

i) Calculate the rated reactive power capacity required at each end as a percentage of the rated active power capacity.

[3]

ii) What is the maximum power that can be transmitted through the HVDC link if the reactive power provision at both ends is limited to 60% of the maximum power transfer?

[1]

iii) How much power can be transmitted if the stipulated ESCR requirement is relaxed to 3.0 with 60% (of rated active power capacity) reactive power provision at both ends?

[1]

2. a) A monopole point-to-point LCC HVDC link has one 6-pulse converter at each end. A proportional controller with gain $K_{pr} = K_{pi} = 5000$ is used for controlling the direct current (I_d) for both converters as shown in Figure 2.1 where, V_{dor} and V_{doi} are the no-load ideal direct voltage at rectifier and inverter end, respectively.

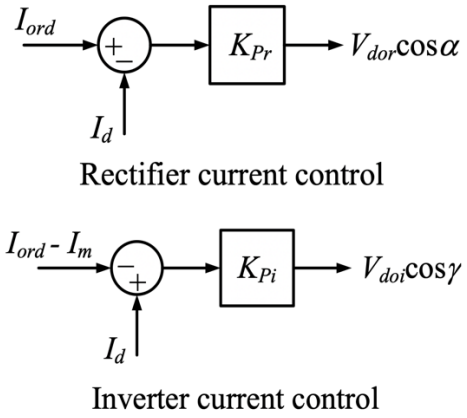


Figure 2.1: Current control scheme for rectifier and inverter

The commutation resistance at both ends is $R_c = 5\ \Omega$ while the DC line resistance is $R_L = 1\ \Omega$. Under normal condition, the AC side voltage at the rectifier and inverter ends are 405 kV and 398 kV, respectively. The turns ratio of the converter transformers at both ends is 0.5 with the low voltage winding on the converter side. The rectifier controls the direct current with a current order $I_{ord} = 1.5$ kA while the inverter operates at a fixed extinction advance angle $\gamma = 18°$. The minimum permissible firing angle for the rectifier is $\alpha_{min} = 5°$. The margin current is $I_m = 250$ A. Calculate the direct current (in kA) through the HVDC link under the following conditions:

i) normal condition.

[4]

ii) if the AC voltage at the rectifier end reduces by 4%.

[2]

iii) if the AC voltage at the rectifier reduces by 6%.

[4]

iv) if the AC voltage at the inverter end reduces by 5% with the rectifier end voltage AC voltage remaining the same as in normal condition.

[2]

b) Two separate AC systems are interconnected by a bipole point-to-point HVDC link using a standard 12-pulse converter arrangement for each pole at both ends. The parameters are given in Table 2.1 below:

Table 2.1: Parameters of the HVDC link

Rated power	$P_{dc-rated}$	1.5 GW
Rated direct voltage	$V_{dc-rated}$	$\pm$400 kV
Resistance of each pole	R_L	2.5 Ω
Minimum firing angle	α_{min}	4°
Minimum extinction advance angle	γ_{min}	16°
Commutation resistance per 6-pulse converter	R_c	5 Ω
Current margin	I_m	20%

At a certain point in time, the AC voltages at the rectifier and inverter end are 240 kV and 275 kV, respectively, and the current control is exercised by the inverter. The transformer tap position is set to provide an effective turns ratio $T = 0.63$ with the low voltage winding on the converter side. Under this condition, determine the following using the calculations on a per pole basis:

i) Direct voltage at the rectifier end.

[3]

ii) Total reactive power consumed by the rectifier.

[1]

iii) Direct voltage at the inverter end.

[1]

iv) Extinction advance angle (in degrees) of the inverter.

[2]

v) Total reactive power consumed by the inverter.

[1]

vi) What effective turns ratio T should be set for the transformers at each end to maintain the rated direct voltage at the rectifier end with minimum reactive power consumption at the inverter end.

[5]

3. a) Answer the following questions related to the DC-DC converter shown in Figure 3.1. The converter parameters are summarised in Table 3.1:

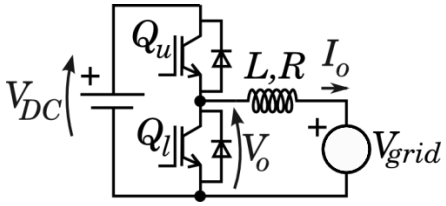


Figure 3.1 – Diagram of a DC-DC converter connected to an ideal DC grid

Parameter	Value	Units	Brief description
V_{grid}	50	V	Nominal DC grid voltage
V_{DC}	100	V	Rated DC link voltage
P_{rated}	500	W	Rated active power delivered to the grid
f_{sw}	20	kHz	Rated switching frequency

Table 3.1 – Characteristic parameters of the DC-DC converter

(i) Write the linear ordinary differential equation (ode) that relates $V_o(t)$, $I_o(t)$ and $V_{grid}(t)$. [2]

(ii) Q_u and Q_l are controlled using PWM with duty cycle D. Sketch the approximate waveform of V_o for a complete switching cycle and give the relationship between the average of V_o, D and V_{DC}.

[1]

(iii) Calculate the rated DC current and the value of the effective series resistance of the filter, R, if we are told that its conduction losses are about 1% of the rated power.

[2]

(iv) Calculate the minimum inductance L that gives a peak-to-peak current ripple of 0.5 A when the output power is of 5 A.

[4]

b) Answer the following questions related to the AC-DC converter shown in Figure 3.2. The converter parameters are summarised in Table 3.2.:

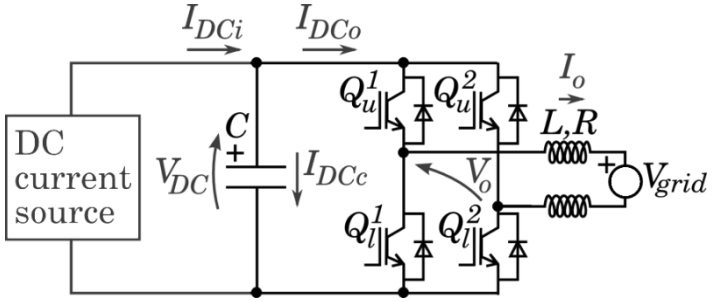


Figure 3.2 – Diagram of a single phase inverter connected to an AC grid

Parameter	Value	Units	Brief description
V_{grid}	230	V rms	Rated AC grid voltage
V_{DC}	400	V	DC link voltage
P_{rated}	2	kW	Rated active power delivered to the grid
f_{rated}	50	Hz	Rated frequency of the AC grid

R	0.12	Ω	Equivalent resistance of bridge plus the filter
L	15	mH	Inductance of the filter

Table 3.2 – Diagram of a single phase inverter connected to an AC grid

(i) Calculate the current required to deliver rated power to the grid at unity power factor. Calculate the losses of the filter and its reactive power consumption.

[1]

(ii) Consider an operating point where the reactive power measured at the bridge terminals (ie at V_o) is 0 and the active power is of 1 kW (towards the AC grid) and answer the following questions neglecting any energy losses:

- What is the minimum V_{DC} required by the converter to generate a V_o of 230 V rms? What is the DC component of I_{DCo} at 1kW if V_{DC} is of 400V?

[2]

- What is the amplitude of the AC component of I_{DCc} caused by the interaction of the 50 Hz voltage and 50 Hz current? How big should the capacitance of the DC link be for the voltage ripple caused by this current to be less than 2V peak-to-peak?

[3]

(iii) Consider the following closed-loop transfer function of a current controller of the inverter where I_o* is the reference current. Answer the questions listed below:

$$I_o(s) = \underbrace{\frac{1}{\tau s + 1}}_{T(s)} I_o^*(s) - \underbrace{\frac{\tau s}{(Ls + R)(\tau s + 1)}}_{SGd(s)} V_{grid}(s)$$

- Considering τ=0.3 ms, what is the gain and phase delay between I_o*(s) and I_o(s) at 50 Hz?

[3]

- Determine the proportional gain, K_P, and the integral gain, K_I, of a PI controller that give the closed-loop transfer function, $T(s)$, shown above.

[2]

c) Answer the following questions related to the AC-DC converter shown in Figure 3.3. The converter parameters are summarised in Table 3.3.:

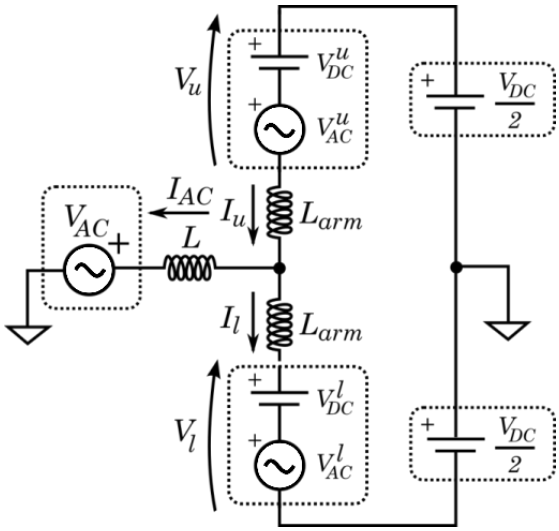


Figure 3.3 – Equivalent circuit of a two-arm single-phase AC/DC modular multilevel converter

Parameter	Value	Description
V_{DC}	1500 V	DC network voltage.
V_{AC}	398 V	AC network voltage (RMS).
f_{AC}	50 Hz	Frequency of the AC network
S_{rated}	300 kVA	Rated power.
X_{arm}	0.11 Ω	Arm reactor inductance.
X_L	0.03 Ω	Phase reactor inductance
R_{arm}, R_L	0	(Negligible) equivalent series resistances

Table 3.3 – Characteristic parameters of the single-phase modular multilevel converter

(i) Calculate the AC and DC components of V_u, V_l, I_u and I_l when delivering rated power to the AC grid at unity power factor (P_{AC}=300 kW, Q_{AC}=0) under long-term stable operation.

[5]

(ii) We are told that the currents of the arms of the converter are: $\bar{I}_{AC}^u$=42 A, I_{DC}^u=35 A, $\bar{I}_{AC}^l$=-38 A, $I_{DC}^l = 35\,A$. Calculate the AC and DC components of the summative current and the current going to the AC grid. Explain briefly the role of the summative current in the operation of the converter.

[3]

(iii) Consider the output voltage of one of the arms (ie $V_{AC}{}^u$ and $V_{DC}{}^u$) to be known and given. What condition must the current of this arm (ie $I_{AC}{}^u$ and $I_{DC}{}^u$) meet in order to prevent the internal energy of the arm from drifting over time?

[2]